LAPORAN JOBSHEET
“19 - Implementasi Unit Testing pada Next.js menggunakan Jest”
MATA KULIAH PEMROGRAMAN BERBASIS FRAMEWORK

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JURUSAN TEKNOLOGI INFORMASI

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PRAKTIKUM 1 – Setup Jest di Next.js

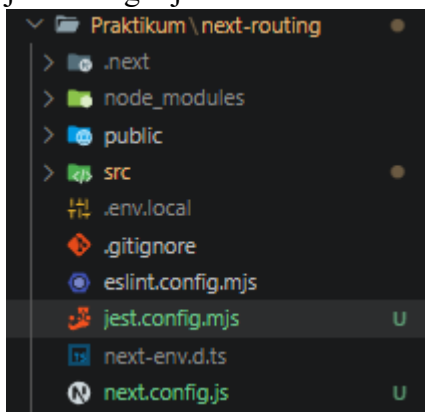
1. Install Dependencies
   Jalankan:npm install jest jest-environment-jsdom @testing-library/react @testinglibrary/jest-dom --save-dev –force

```
PS D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing> npm install jest jest-environment-jsdom @testing-library/react @testing-library/jest-dom --save-
dev --force
npm warn using --force Recommended protections disabled.
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want a good and tested way to coalesce async request
s by a key value, which is much more comprehensive and powerful.
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Plea
se update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me
npm warn deprecated whatwg-encoding@3.1.1: Use @exodus/bytes instead for a more spec-conformant and faster implementation
npm warn deprecated glob@10.5.0: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Ple
ase update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me

added 286 packages, and audited 780 packages in 26s

195 packages are looking for funding
  run `npm fund` for details

5 vulnerabilities (3 moderate, 1 high, 1 critical)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
```

   Generate basic Jest Configuration

```
PS D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing> npm init jest@latest
Need to install the following packages:
create-jest@30.3.0
Ok to proceed? (y) y

npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want a good and tested way to coalesce async request
s by a key value, which is much more comprehensive and powerful.
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Plea
se update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me
npm warn deprecated glob@10.5.0: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Ple
ase update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me

> next-routing@0.1.0 npx
> create-jest


The following questions will help Jest to create a suitable configuration for your project

√ Would you like to use Jest when running "test" script in "package.json"? ... yes
√ Would you like to use Typescript for the configuration file? ... yes
√ Choose the test environment that will be used for testing » jsdom (browser-like)
√ Do you want Jest to add coverage reports? ... yes
√ Which provider should be used to instrument code for coverage? » v8
√ Automatically clear mock calls, instances, contexts and results before every test? ... yes

 Modified D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing\package.json

 Configuration file created at D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing\jest.config.ts
```

2. Buat File Konfigurasi
   Doc : https://nextjs.org/docs/pages/guides/testing/jest
   - Buat file
     jest.config.mjs



- Isi dengan:

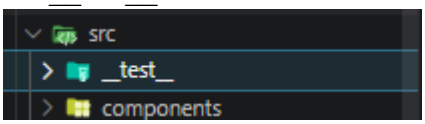
```
P2 > Praktikum > next-routing > jest.config.mjs > default
import nextJest from 'next/jest.js'

const createJestConfig = nextJest({
  dir: './',
})

const config = {
  coverageProvider: 'v8',
  testEnvironment: 'jsdom',
}

export default createJestConfig(config)
```

3. Tambahkan Script di package.json

```
  "name": "next-routing",
  "version": "0.1.0",
  "private": true,
  ▷Debug
  "scripts": {
    "dev": "next dev",
    "build": "next build",
    "start": "next start",
    "lint": "eslint",
    "test": "jest --passWithNoTests -u",
    "test:coverage": "npm run test -- --coverage",
    "test:watch": "jest --watch"
```

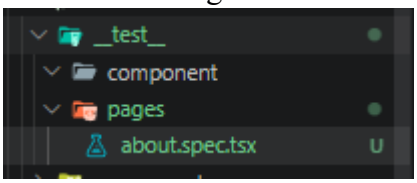
PRAKTIKUM 2 – Struktur Folder Testing

1. Buat folder:
   - src/__test__/



PRAKTIKUM 3 – Testing Halaman About

1. Buat File Testing



2. Contoh Testing Snapshot. Pada about.spec.tsx tambahkan code berikut :

```
P2 > Praktikum > next-routing > src > __test__ > pages > about.spec.tsx >
import {render} from "@testing-library/react";
import AboutPage from "@/pages/about";

describe("AboutPage", () => {
    it("renders the about page correctly", () => {
        const page = render(<AboutPage />);
        expect(page).toMatchSnapshot();
    });
})
```

3. Jalankan Testing
   - npm run test
   - Jika berhasil:
   - PASS about.spec.tsx

```
PS D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing> npm run test

> next-routing@0.1.0 test
> jest --passWithNoTests -u

 PASS  src/__test__/pages/about.spec.tsx
  AboutPage
    √ renders the about page correctly (35 ms)

 › 1 snapshot written.
Snapshot Summary
 › 1 snapshot written from 1 test suite.

Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   1 written, 1 total
Time:        3.418 s
```

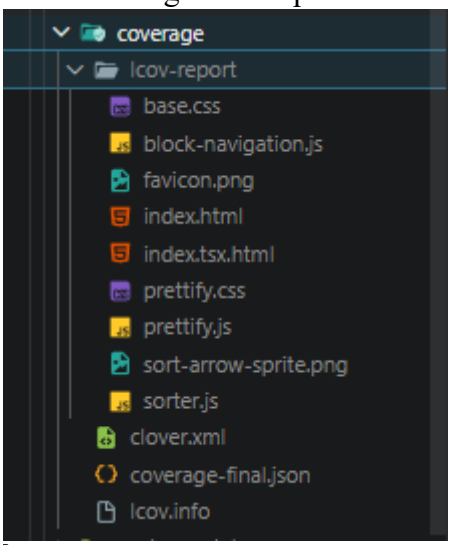
PRAKTIKUM 4 – Coverage Report

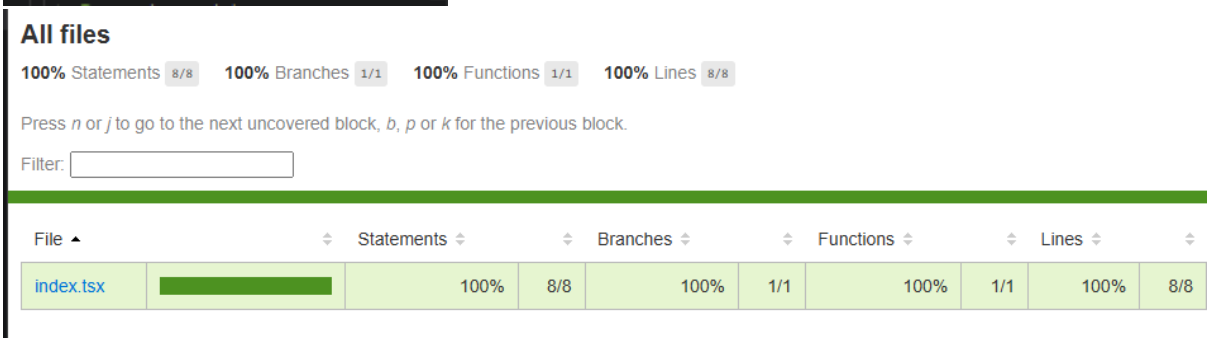
1. Jalankan:
   - npm run test:coverage

```
PS D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing> npm run test:coverage

> next-routing@0.1.0 test:coverage
> npm run test -- --coverage


> next-routing@0.1.0 test
> jest --passWithNoTests -u --coverage

 PASS  src/__test__/pages/about.spec.tsx
  AboutPage
    √ renders the about page correctly (41 ms)

-----------|---------|----------|---------|---------|-------------------
File       | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-----------|---------|----------|---------|---------|-------------------
All files  |     100 |      100 |     100 |     100 |
 index.tsx |     100 |      100 |     100 |     100 |
-----------|---------|----------|---------|---------|-------------------
Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   1 passed, 1 total
Time:        4.033 s
Ran all test suites.
```

2. Akan muncul folder:
   - coverage/

- Buka coverage/lcov-report/index.html (buka di melalui explorer)





PRAKTIKUM 5 – Konfigurasi Coverage Lengkap

1. Update jest.config.mjs:

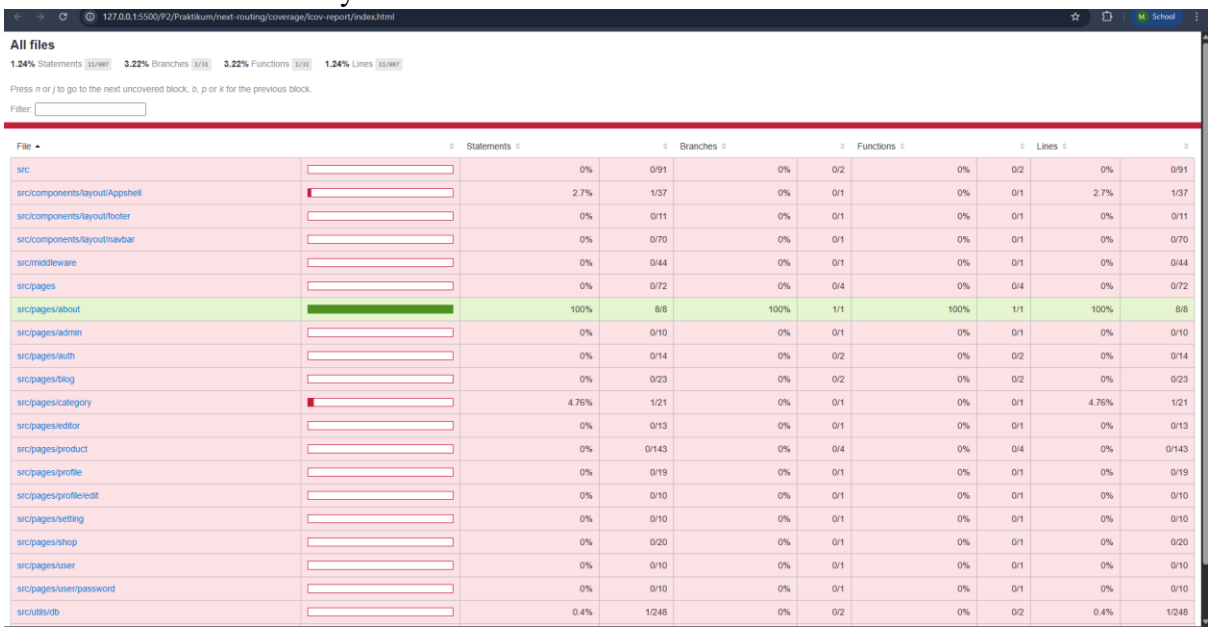
P2 > Praktikum > next-routing > jest.config.mjs > default

```
import nextJest from 'next/jest.js'

const createJestConfig = nextJest({
  dir: './',
})

const config = {
  coverageProvider: 'v8',
  testEnvironment: 'jsdom',
  modulePaths: ['<rootDir>/src/'],
  collectCoverage: true,
  collectCoverageFrom: [
    'src/**/*.{ts,tsx}',
    'src/**/*.d.ts',
    "!**/node_modules/**",
    "!**/.next/**",
    "!**/coverage/**",
    "!**/jest.config.mjs",
    "!**/next.config.mjs",
    "!**/src/types/**",
    "!**/views/**",
    "!**/pages/api/**",
  ]
}

export default createJestConfig(config)
```

2. Jalankan npm run test:coverage

```
PS D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing> npm run test:coverage

> next-routing@0.1.0 test:coverage
> npm run test -- --coverage


> next-routing@0.1.0 test
> jest --passWithNoTests -u --coverage

 PASS  src/__test__/pages/about.spec.tsx
  AboutPage
    √ renders the about page correctly (41 ms)

-------------------------------|---------|----------|---------|---------|-------------------
File                           | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------------------|---------|----------|---------|---------|-------------------
All files                      |    1.24 |     3.22 |    3.22 |    1.24 |
 src                           |       0 |        0 |       0 |       0 |
  middleware.ts                |       0 |        0 |       0 |       0 | 1-87
  styles.d.ts                  |       0 |        0 |       0 |       0 | 1-4
 src/components/layout/Appshell |     2.7 |        0 |       0 |     2.7 |
  index.tsx                    |     2.7 |        0 |       0 |     2.7 | 1-36
 src/components/layout/footer  |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-11
 src/components/layout/navbar  |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-70
 src/middleware                |       0 |        0 |       0 |       0 |
  withAuth.ts                  |       0 |        0 |       0 |       0 | 1-44
 src/pages                     |       0 |        0 |       0 |       0 |
  404.tsx                      |       0 |        0 |       0 |       0 | 1-32
  _app.tsx                     |       0 |        0 |       0 |       0 | 1-14
  _document.tsx                |       0 |        0 |       0 |       0 | 1-13
  index.tsx                    |       0 |        0 |       0 |       0 | 1-13
 src/pages/about               |     100 |      100 |     100 |     100 |
  index.tsx                    |     100 |      100 |     100 |     100 |
 src/pages/admin               |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-10
 src/pages/auth                |       0 |        0 |       0 |       0 |
  login.tsx                    |       0 |        0 |       0 |       0 | 1-7
  register.tsx                 |       0 |        0 |       0 |       0 | 1-7
 src/pages/blog                |       0 |        0 |       0 |       0 |
  [slug].tsx                   |       0 |        0 |       0 |       0 | 1-13
  index.tsx                    |       0 |        0 |       0 |       0 | 1-10
 src/pages/category            |    4.76 |        0 |       0 |    4.76 |
  [[...slug]].tsx              |    4.76 |        0 |       0 |    4.76 | 1-20
 src/pages/editor              |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-13
 src/pages/product             |       0 |        0 |       0 |       0 |
  [product].tsx                |       0 |        0 |       0 |       0 | 1-68
  index.tsx                    |       0 |        0 |       0 |       0 | 1-16
  server.tsx                   |       0 |        0 |       0 |       0 | 1-29
  static.tsx                   |       0 |        0 |       0 |       0 | 1-30
 src/pages/profile             |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-19
 src/pages/profile/edit        |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-10
```

3. Jika dilihat di index.htmlnya



File		Statements		Branches		Functions		Lines	
src		0%	0/91	0%	0/2	0%	0/2	0%	0/91
src/components/layout/Appshell		2.7%	1/37	0%	0/1	0%	0/1	2.7%	1/37
src/components/layout/footer		0%	0/11	0%	0/1	0%	0/1	0%	0/11
src/components/layout/navbar		0%	0/70	0%	0/1	0%	0/1	0%	0/70
src/middleware		0%	0/44	0%	0/1	0%	0/1	0%	0/44
src/pages		0%	0/72	0%	0/4	0%	0/4	0%	0/72
src/pages/about		100%	8/8	100%	1/1	100%	1/1	100%	8/8
src/pages/admin		0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/auth		0%	0/14	0%	0/2	0%	0/2	0%	0/14
src/pages/blog		0%	0/23	0%	0/2	0%	0/2	0%	0/23
src/pages/category		4.76%	1/21	0%	0/1	0%	0/1	4.76%	1/21
src/pages/editor		0%	0/13	0%	0/1	0%	0/1	0%	0/13
src/pages/product		0%	0/143	0%	0/4	0%	0/4	0%	0/143
src/pages/profile		0%	0/19	0%	0/1	0%	0/1	0%	0/19
src/pages/profile/edit		0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/setting		0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/shop		0%	0/20	0%	0/1	0%	0/1	0%	0/20
src/pages/user		0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/user/password		0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/utils/db		0.4%	1/248	0%	0/2	0%	0/2	0.4%	1/248

PRAKTIKUM 6 – Testing dengan getByTestId

1. Tambahkan pada About Page
   - <h1 data-testid="title">About Page</h1>

P2 > Praktikum > next-routing > src > pages > about > index.tsx > AboutPage

```
const AboutPage = () => {
    return (
        <div>
            <h1 data-testid="title">About Page</h1>
        </div>
    );
};

export default AboutPage;
```

2. Update Testing pada about.spec.tsx

xt-routing > src > __test__ > pages > about.spec.tsx > describe("AboutPage") callback > it("renders t

```
import {render} from "@testing-library/react";
import AboutPage from "@/pages/about";

describe("AboutPage", () => {
    it("renders the about page correctly", () => {
        const page = render(<AboutPage />);
        expect(page.getByTestId("title").textContent).toBe("About Page");
        expect(page).toMatchSnapshot();
    });
})
```

3. Dicoba untuk di run

```
PS D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing> npm run test:coverage

> next-routing@0.1.0 test:coverage
> npm run test -- --coverage


> next-routing@0.1.0 test
> jest --passWithNoTests -u --coverage

PASS  src/__test__/pages/about.spec.tsx
  AboutPage
    √ renders the about page correctly (53 ms)

 › 1 snapshot updated.
--------------------------------|---------|----------|---------|---------|-------------------
File                            | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
--------------------------------|---------|----------|---------|---------|-------------------
All files                       |    1.35 |     3.22 |    3.22 |    1.35 |
 src                            |       0 |        0 |       0 |       0 |
  middleware.ts                 |       0 |        0 |       0 |       0 | 1-87
  styles.d.ts                   |       0 |        0 |       0 |       0 | 1-4
 src/components/layout/Appshell |     2.7 |        0 |       0 |     2.7 |
  index.tsx                     |     2.7 |        0 |       0 |     2.7 | 1-36
 src/components/layout/footer   |       0 |        0 |       0 |       0 |
  index.tsx                     |       0 |        0 |       0 |       0 | 1-11
 src/components/layout/navbar   |       0 |        0 |       0 |       0 |
  index.tsx                     |       0 |        0 |       0 |       0 | 1-70
 src/middleware                 |       0 |        0 |       0 |       0 |
  withAuth.ts                   |       0 |        0 |       0 |       0 | 1-44
 src/pages                      |       0 |        0 |       0 |       0 |
  404.tsx                       |       0 |        0 |       0 |       0 | 1-32
  _app.tsx                      |       0 |        0 |       0 |       0 | 1-14
  _document.tsx                 |       0 |        0 |       0 |       0 | 1-13
  index.tsx                     |       0 |        0 |       0 |       0 | 1-13
 src/pages/about                |     100 |      100 |     100 |     100 |
  index.tsx                     |     100 |      100 |     100 |     100 |
 src/pages/admin                |       0 |        0 |       0 |       0 |
  index.tsx                     |       0 |        0 |       0 |       0 | 1-10
 src/pages/auth                 |       0 |        0 |       0 |       0 |
```

4. Coba Jika dibuat Salah:
   - Rubah menjadi toBe("About")

```
P2 > Praktikum > next-routing > src > __test__ > pages > about.spec.tsx > describe("AboutPage
1   import {render} from "@testing-library/react";
2   import AboutPage from "@/pages/about";
3
4   describe("AboutPage", () => {
5       it("renders the about page correctly", () => {
6           const page = render(<AboutPage />);
7           expect(page.getByTestId("title").textContent).toBe("About");
8           expect(page).toMatchSnapshot();
9       });
10  })
```

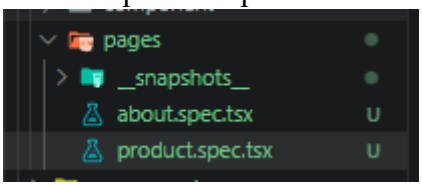
5. Jalankan dan Hasil:
   - FAIL Expected: "About"
   - Received: "About Page"

```
PS D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing> npm run test:coverage

  > next-routing@0.1.0 test:coverage
  > npm run test -- --coverage


  > next-routing@0.1.0 test
  > jest --passWithNoTests -u --coverage

  FAIL src/__test__/pages/about.spec.tsx
    AboutPage
      × renders the about page correctly (45 ms)

    ● AboutPage › renders the about page correctly

      expect(received).toBe(expected) // Object.is equality

      Expected: "About"
      Received: "About Page"

        5 |     it("renders the about page correctly", () => {
        6 |         const page = render(<AboutPage />);
      > 7 |         expect(page.getByTestId("title").textContent).toBe("About");
          |                                                       ^
        8 |         expect(page).toMatchSnapshot();
        9 |     });
       10 | })

      at Object.toBe (src/__test__/pages/about.spec.tsx:7:55)
-------------------------------|---------|------
Test Suites: 1 failed, 1 total
Tests:       1 failed, 1 total
Snapshots:   0 total
Time:        20.435 s
Ran all test suites.
```

PRAKTIKUM 7 – Testing Page dengan Router (Mocking)

Kita coba untuk melakukan testing pada halaman produk

1. Buat file product.spec.tsx



2. Tambahkan kode berikut

```
P2 > Praktikum > next-routing > src > __test__ > pages > product.spec.tsx > ...
import { render, screen } from "@testing-library/react";
import ProductPage from "@/pages/product";

describe("ProductPage", () => {
    it("renders the product page correctly", () => {
        const page = render(<ProductPage />);
        expect(page.getByTestId("title").textContent).toBe("Product Page");
        expect(page).toMatchSnapshot();
    });
});
```

3. Ketika testing halaman Product, sering muncul error:
   - NextRouter was not mounted

```
● ProductPage › renders the product page correctly

  TestingLibraryElementError: Unable to find an element by: [data-testid="title"]

  Ignored nodes: comments, script, style
  <body>
    <div>
      <div
        style="padding: 2rem; text-align: center;"
      >
        <p>
          Loading Products...
        </p>
      </div>
    </div>
  </body>

    5 |     it("renders the product page correctly", () => {
    6 |         const page = render(<ProductPage />);
  > 7 |         expect(page.getByTestId("title").textContent).toBe("Product Page");
      |                                                       ^
    8 |         expect(page).toMatchSnapshot();
    9 |     });
   10 | });

    at Object.getElementError (node_modules/@testing-library/dom/dist/config.js:37:19)
    at node_modules/@testing-library/dom/dist/query-helpers.js:76:38
    at node_modules/@testing-library/dom/dist/query-helpers.js:52:17
    at node_modules/@testing-library/dom/dist/query-helpers.js:95:19
    at Object.getByTestId (src/__test__/pages/product.spec.tsx:7:21)

● ProductPage › renders the product page correctly

  NextRouter was not mounted. https://nextjs.org/docs/messages/next-router-not-mounted
```

Solusi: Mock Next Router

- Tambahkan di file product.spec.tsx

```
P2 > Praktikum > next-routing > src > __test__ > pages > product.spec.tsx > jest.mock("next/router") callback >
import { render, screen } from "@testing-library/react";
import ProductPage from "@/pages/product";

jest.mock("next/router", () => ({
  useRouter() {
    return {
      route: "/product",
      pathname: "",
      query: "",
      asPath: "",
      push: jest.fn(),
      events: {
        on: jest.fn(),
        off: jest.fn(),
      },
      isReady: true,
    };
  },
}));

describe("ProductPage", () => {
  it("renders the product page correctly", () => {
    const page = render(<ProductPage />);
    expect(page.getByTestId("title").textContent).toBe("Product Page");
    expect(page).toMatchSnapshot();
  });
});
```

PRAKTIKUM 8 – Menangani Undefined Data

1. Jalankan npm run test:coverage maka akan muncul error

```
● ProductPage › renders the product page correctly

  TestingLibraryElementError: Unable to find an element by: [data-testid="title"]

  Ignored nodes: comments, script, style
  <body>
    <div>
      <div
        style="padding: 2rem; text-align: center;"
      >
        <p>
          Loading Products...
        </p>
      </div>
    </div>
  </body>

    22 |   it("renders the product page correctly", () => {
    23 |     const page = render(<ProductPage />);
  > 24 |     expect(page.getByTestId("title").textContent).toBe("Product Page");
       |                                      ^
    25 |     expect(page).toMatchSnapshot();
    26 |   });
    27 | });

    at Object.getElementError (node_modules/@testing-library/dom/dist/config.js:37:19)
    at node_modules/@testing-library/dom/dist/query-helpers.js:76:38
```

2. Error saya karena, Unable to find an element by: [data-testid="title"]
   Jadi, Jest mencoba mencari elemen dengan atribut data-testid="title", tapi yang ditemukan justru teks "Loading Products...".

3. Fix dengan findByTestId (yang bersifat *asynchronous*) bukan getByTestId

```
describe("ProductPage", () => {
  it("renders the product page correctly", async() => {
    const page = render(<ProductPage />);
    const titleElement = await screen.findByTestId("title");
    expect(titleElement.textContent).toBe("Product Page");
    expect(page).toMatchSnapshot();
  });
});
```

Analisis Coverage

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	3.49	17.64	12.5	3.49	
src	0	0	0	0	
middleware.ts	0	0	0	0	1-87
styles.d.ts	0	0	0	0	1-4
src/components/layout/Appshell	2.7	0	0	2.7	
index.tsx	2.7	0	0	2.7	1-36
src/components/layout/footer	0	0	0	0	
index.tsx	0	0	0	0	1-11
src/components/layout/navbar	0	0	0	0	
index.tsx	0	0	0	0	1-70
src/middleware	0	0	0	0	
withAuth.ts	0	0	0	0	1-44
src/pages	0	0	0	0	
404.tsx	0	0	0	0	1-32
_app.tsx	0	0	0	0	1-14
_document.tsx	0	0	0	0	1-13
index.tsx	0	0	0	0	1-13
src/pages/about	100	100	100	100	
index.tsx	100	100	100	100	
src/pages/admin	0	0	0	0	
index.tsx	0	0	0	0	1-10
src/pages/auth	0	0	0	0	
login.tsx	0	0	0	0	1-7
register.tsx	0	0	0	0	1-7
src/pages/blog	0	0	0	0	
[slug].tsx	0	0	0	0	1-13
index.tsx	0	0	0	0	1-10
src/pages/category	4.76	0	0	4.76	
[[...slug]].tsx	4.76	0	0	4.76	1-20
src/pages/editor	0	0	0	0	
index.tsx	0	0	0	0	1-13
src/pages/product	11.18	57.14	40	11.18	
[product].tsx	0	0	0	0	1-68
index.tsx	100	100	100	100	
server.tsx	0	0	0	0	1-29
static.tsx	0	0	0	0	1-30
src/pages/profile	0	0	0	0	
index.tsx	0	0	0	0	1-19
src/pages/profile/edit	0	0	0	0	
index.tsx	0	0	0	0	1-10
src/pages/setting	0	0	0	0	
app.tsx	0	0	0	0	1-10
src/pages/shop	0	0	0	0	
[[...slug]].tsx	0	0	0	0	1-20
src/pages/user	0	0	0	0	
index.tsx	0	0	0	0	1-10
src/pages/user/password	0	0	0	0	
index.tsx	0	0	0	0	1-10

File		Statements		Branches		Functions		Lines	
src		0%	0/91	0%	0/2	0%	0/2	0%	0/91
src/components/layout/Appshell		2.7%	1/37	0%	0/1	0%	0/1	2.7%	1/37
src/components/layout/footer		0%	0/11	0%	0/1	0%	0/1	0%	0/11
src/components/layout/navbar		0%	0/70	0%	0/1	0%	0/1	0%	0/70
src/middleware		0%	0/44	0%	0/1	0%	0/1	0%	0/44
src/pages		0%	0/72	0%	0/4	0%	0/4	0%	0/72
src/pages/about		100%	9/9	100%	1/1	100%	1/1	100%	9/9
src/pages/admin		0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/auth		0%	0/14	0%	0/2	0%	0/2	0%	0/14
src/pages/blog		0%	0/23	0%	0/2	0%	0/2	0%	0/23
src/pages/category		4.76%	1/21	0%	0/1	0%	0/1	4.76%	1/21
src/pages/editor		0%	0/13	0%	0/1	0%	0/1	0%	0/13
src/pages/product		11.18%	16/143	57.14%	4/7	40%	2/5	11.18%	16/143
src/pages/profile		0%	0/19	0%	0/1	0%	0/1	0%	0/19
src/pages/profile/edit		0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/setting		0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/shop		0%	0/20	0%	0/1	0%	0/1	0%	0/20
src/pages/user		0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/user/password		0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/utils/db		0.4%	1/248	0%	0/2	0%	0/2	0.4%	1/248
src/utils/swr		100%	3/3	100%	1/1	100%	1/1	100%	3/3

Jika dilihat aplikasi yang dibuat masih 5% jadi perlu banyak perbaikan sebelum proses deploy.

Perhatikan bagian:

- Statement
- Branch
- Function
- Lines

Contoh: Metric dan Hasil

- Statements 85%
- Branch 60%
- Functions 90%
- Lines 88%

Branch biasanya paling sulit karena perlu menguji kondisi if/else

tandar Industri Biasanya:

- ≥ 80% → boleh production
- 80% → harus diperbaiki
- Semua critical feature wajib dites

TUGAS PRAKTIKUM

1. Buat unit test untuk:
   - Halaman Product
   - 1 Komponen

P2 > Praktikum > next-routing > src > __test__ > pages > product.spec.tsx > jest.mock("next/router")

```
import { render, screen } from "@testing-library/react";
import ProductPage from "@/pages/product";
import "@testing-library/jest-dom";

jest.mock("next/router", () => ({
  useRouter() {
    return {
      route: "/product",
      pathname: "",
      query: "",
      asPath: "",
      push: jest.fn(),
      events: {
        on: jest.fn(),
        off: jest.fn(),
      },
      isReady: true,
    };
  },
}));

describe("ProductPage", () => {
  it("renders the product page correctly", async () => {
    const page = render(<ProductPage />);
    const titleElement = await screen.findByRole("heading", {
      name: /Daftar Produk/i,
    });
    expect(titleElement.textContent).toBe("Daftar Produk");
    expect(page).toMatchSnapshot();
  });

  it("renders the product page container with correct testid", () => {
    render(<ProductPage />);
    const container = screen.getByTestId("product-page-container");
    expect(container).toBeInTheDocument();
  });
});
```

2. Gunakan minimal:
   - 1 Snapshot test

```
expect(titleElement.textContent).toBe("Daftar Produk");
expect(page).toMatchSnapshot();
```

   - 1 toBe()

```
expect(titleElement.textContent).toBe("Daftar Produk");
expect(page).toMatchSnapshot();
```

```
expect(container).toBeInTheDocument();
```

   - 1 getByTestId()

```
const container = screen.getByTestId("product-page-container");
expect(container).toBeInTheDocument();
```

3. Buat coverage minimal 50%
   AbuoutPage
   Saya hanya testing getByTestId

```
P2 > Praktikum > next-routing > src > __test__ > pages > about.spec.tsx > ...
import {render} from "@testing-library/react";
import AboutPage from "@/pages/about";

describe("AboutPage", () => {
    it("renders the about page correctly", () => {
        const page = render(<AboutPage />);
        expect(page.getByTestId("title").textContent).toBe("About Page");
        expect(page).toMatchSnapshot();
    });
})
```

Yang hasilnya:

All files

3.49% Statements 31/888 **15.15%** Branches 5/33 **9.37%** Functions 3/32 **3.49%** Lines 31/888

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
src		0%	0/91	0%	0/2	0%	0/2	0%	0/91
src/components/layout/Appshell		2.7%	1/37	0%	0/1	0%	0/1	2.7%	1/37
src/components/layout/footer		0%	0/11	0%	0/1	0%	0/1	0%	0/11
src/components/layout/navbar		0%	0/70	0%	0/1	0%	0/1	0%	0/70
src/middleware		0%	0/44	0%	0/1	0%	0/1	0%	0/44
src/pages		0%	0/72	0%	0/4	0%	0/4	0%	0/72
src/pages/about		100%	9/9	100%	1/1	100%	1/1	100%	9/9
src/pages/admin		0%	0/10	0%	0/1	0%	0/1	0%	0/10

100% di AboutPage terilhat di metrik statement, branches, function, maupun lines karena isinya hanya index.tsx

a. ProductPage
   - Pengujian Main (Heading & Container)

```
it("renders the product page correctly", async () => {
  const page = render(<ProductPage />);
  const titleElement = await screen.findByRole("heading", {
    name: /Daftar Produk/i,
  });
  expect(titleElement.textContent).toBe("Daftar Produk");
  expect(page).toMatchSnapshot();
});

it("renders the product page container with correct testid", () => {
  render(<ProductPage />);
  const container = screen.getByTestId("product-page-container");
  expect(container).toBeInTheDocument();
});
```

Penjelasan:

1. renders the product page correctly: Tes ini memastikan judul utama "Daftar Produk" muncul. Penggunaan findByRole sangat bagus karena ia menunggu proses asynchronous selesai.
2. Snapshot Test: expect(page).toMatchSnapshot() mengambil "foto" dari struktur HTML halaman Anda untuk mendeteksi jika ada perubahan UI yang tidak disengaja di masa depan.
3. renders product page container: Memastikan elemen pembungkus utama memiliki atribut data-testid="product-page-container". Ini adalah cara paling aman untuk menargetkan elemen di Testing Library.

- Pengujian Styling dan Aset

```
it("should render product page with correct class", () => {
  render(<ProductPage />);
  const container = screen.getByTestId("product-page-container");
  expect(container).toHaveClass("min-h-screen");
});

it("should render hero section", async () => {
  render(<ProductPage />);
  const images = await screen.findAllByAltText("Product Icon");
  expect(images.length).toBeGreaterThan(0);
});
```

Penjelasan:

1. should render product page with correct class: Anda menguji apakah CSS Tailwind min-h-screen terpasang. Ini memastikan layouting halaman tetap konsisten.
2. should render hero section: Anda menggunakan findAllByAltText untuk mengecek keberadaan gambar (ikon produk). Ini memastikan tag <img> memiliki deskripsi alt yang benar untuk aksesibilitas.

- Pengujian Interaksi dan Konten Teks

```
it("should render logout button", async () => {
  render(<ProductPage />);
  const logoutButton = await screen.findByRole("button", {
    name: /Logout/i,
  });
  expect(logoutButton).toBeInTheDocument();
});

it("should render welcome message", async () => {
  render(<ProductPage />);
  const welcomeText = await screen.findByText(/Selamat datang/i);
  expect(welcomeText).toBeInTheDocument();
});

it("should render subtitle text", async () => {
  render(<ProductPage />);
  const subtitleText = await screen.findByText(/Jelajahi berbagai produk/i);
  expect(subtitleText).toBeInTheDocument();
});

it("should render refresh button", async () => {
  render(<ProductPage />);
  const refreshButton = await screen.findByRole("button", {
    name: /Refresh/i,
  });
  expect(refreshButton).toBeInTheDocument();
});
```

Penjelasan:

1. Tombol (Logout & Refresh): Anda menggunakan findByRole("button", { name: /.../i }). Ini sangat direkomendasikan karena menguji fungsionalitas sekaligus aksesibilitas (memastikan tombol memiliki label yang terbaca oleh screen reader).
2. Teks Selamat Datang & Subtitle: Menggunakan findByText dengan regex (/i) agar pencarian teks tidak sensitif terhadap huruf besar/kecil. Ini memastikan pesan pemasaran atau sapaan kepada user sudah muncul.

- Pengujian Struktur (Head & Headings)

```
  it("should render page title in head", async () => {
    render(<ProductPage />);
    // This tests that the component renders without errors
    const container = screen.getByTestId("product-page-container");
    expect(container.parentElement).toBeTruthy();
  });

  it("should have multiple headings on page", async () => {
    render(<ProductPage />);
    const headings = screen.getAllByRole("heading");
    expect(headings.length).toBeGreaterThanOrEqual(2);
  });
});
```

Penjelasan:

1. Multiple Headings: Tes ini memastikan halaman memiliki struktur hierarki yang baik (misalnya ada <h1> untuk judul dan <h2> untuk sub-judul) dengan mengecek apakah jumlah elemen heading lebih dari atau sama dengan dua.

4. Lakukan Mocking untuk router
   - Mocking Route pada ProductPage

```
// Mock next/router
jest.mock("next/router", () => ({
  useRouter() {
    return {
      route: "/product",
      pathname: "",
      query: "",
      asPath: "",
      push: jest.fn(),
      events: {
        on: jest.fn(),
        off: jest.fn(),
      },
      isReady: true,
    };
  },
}));
```

- Mocking SWR pada ProductPage

```
// Mock SWR
jest.mock("swr", () => ({
  __esModule: true,
  default: jest.fn(() => ({
    data: { data: [] },
    error: null,
    isLoading: false,
    isValidating: false,
    mutate: jest.fn(),
  })),
}));
```

5. Dokumentasikan hasil coverage

```
A worker process has failed to exit gracefully and has been force exited. This is likely caused by tests leaking due to impr
-------------------------------|---------|----------|---------|---------|-------------------
File                           | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------------------|---------|----------|---------|---------|-------------------
All files                      |    3.49 |    15.15 |    9.37 |    3.49 |
 src                           |       0 |        0 |       0 |       0 |
  middleware.ts                |       0 |        0 |       0 |       0 | 1-87
  styles.d.ts                  |       0 |        0 |       0 |       0 | 1-4
 src/components/layout/Appshell |     2.7 |        0 |       0 |     2.7 |
  index.tsx                    |     2.7 |        0 |       0 |     2.7 | 1-36
 src/components/layout/footer  |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-11
 src/components/layout/navbar  |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-70
 src/middleware                |       0 |        0 |       0 |       0 |
  withAuth.ts                  |       0 |        0 |       0 |       0 | 1-44
 src/pages                     |       0 |        0 |       0 |       0 |
  404.tsx                      |       0 |        0 |       0 |       0 | 1-32
  _app.tsx                     |       0 |        0 |       0 |       0 | 1-14
  _document.tsx                |       0 |        0 |       0 |       0 | 1-13
  index.tsx                    |       0 |        0 |       0 |       0 | 1-13
 src/pages/about               |     100 |      100 |     100 |     100 |
  index.tsx                    |     100 |      100 |     100 |     100 |
 src/pages/admin               |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-10
 src/pages/auth                |       0 |        0 |       0 |       0 |
  login.tsx                    |       0 |        0 |       0 |       0 | 1-7
  register.tsx                 |       0 |        0 |       0 |       0 | 1-7
 src/pages/blog                |       0 |        0 |       0 |       0 |
  [slug].tsx                   |       0 |        0 |       0 |       0 | 1-13
  index.tsx                    |       0 |        0 |       0 |       0 | 1-10
 src/pages/category            |    4.76 |        0 |       0 |    4.76 |
  [[...slug]].tsx              |    4.76 |        0 |       0 |    4.76 | 1-20
 src/pages/editor              |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-13
 src/pages/product             |   11.18 |    57.14 |      40 |   11.18 |
  [product].tsx                |       0 |        0 |       0 |       0 | 1-68
  index.tsx                    |     100 |      100 |     100 |     100 |
  server.tsx                   |       0 |        0 |       0 |       0 | 1-29
  static.tsx                   |       0 |        0 |       0 |       0 | 1-30
 src/pages/profile             |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-19
 src/pages/profile/edit        |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-10
 src/pages/setting             |       0 |        0 |       0 |       0 |
  app.tsx                      |       0 |        0 |       0 |       0 | 1-10
 src/pages/shop                |       0 |        0 |       0 |       0 |
  [[...slug]].tsx              |       0 |        0 |       0 |       0 | 1-20
 src/pages/user                |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-10
 src/pages/user/password       |       0 |        0 |       0 |       0 |
  index.tsx                    |       0 |        0 |       0 |       0 | 1-10
 src/utils/db                  |     0.4 |        0 |       0 |     0.4 |
  firebase.ts                  |       0 |        0 |       0 |       0 | 1-17
  servicesfirebase.ts          |    0.43 |        0 |       0 |    0.43 | 1-230
 src/utils/swr                 |     100 |      100 |       0 |     100 |
  fetcher.ts                   |     100 |      100 |       0 |     100 |
-------------------------------|---------|----------|---------|---------|-------------------

Test Suites: 2 passed, 2 total
Tests:       11 passed, 11 total
Snapshots:   2 passed, 2 total
Time:        41.014 s
Ran all test suites.
```

File	Statements		Branches		Functions		Lines	
src	0%	0/91	0%	0/2	0%	0/2	0%	0/91
src/components/layout/Appshell	2.7%	1/37	0%	0/1	0%	0/1	2.7%	1/37
src/components/layout/footer	0%	0/11	0%	0/1	0%	0/1	0%	0/11
src/components/layout/navbar	0%	0/70	0%	0/1	0%	0/1	0%	0/70
src/middleware	0%	0/44	0%	0/1	0%	0/1	0%	0/44
src/pages	0%	0/72	0%	0/4	0%	0/4	0%	0/72
src/pages/about	100%	9/9	100%	1/1	100%	1/1	100%	9/9
src/pages/admin	0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/auth	0%	0/14	0%	0/2	0%	0/2	0%	0/14
src/pages/blog	0%	0/23	0%	0/2	0%	0/2	0%	0/23
src/pages/category	4.76%	1/21	0%	0/1	0%	0/1	4.76%	1/21
src/pages/editor	0%	0/13	0%	0/1	0%	0/1	0%	0/13
src/pages/product	11.18%	16/143	57.14%	4/7	40%	2/5	11.18%	16/143
src/pages/profile	0%	0/19	0%	0/1	0%	0/1	0%	0/19
src/pages/profile/edit	0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/setting	0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/shop	0%	0/20	0%	0/1	0%	0/1	0%	0/20
src/pages/user	0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/pages/user/password	0%	0/10	0%	0/1	0%	0/1	0%	0/10
src/utils/db	0.4%	1/248	0%	0/2	0%	0/2	0.4%	1/248
src/utils/swr	100%	3/3	100%	0/0	0%	0/1	100%	3/3

Pada ProductPage semua metrik lengkap hanya pada index.tsx nya saja, karena saya focus pada testing index.tsx

DISKUSI DAN REFLEKSI

1. **Mengapa unit testing penting sebelum production?**
   Implementasi *unit testing* sebelum tahap produksi berfungsi sebagai mekanisme verifikasi kualitas yang memastikan setiap unit terkecil dari kode, seperti komponen atau fungsi, berjalan sesuai spesifikasi. Secara teknis, pengujian ini krusial untuk mendeteksi *bug* atau kegagalan logika di tahap awal pengembangan, sehingga biaya perbaikan menjadi lebih rendah dibandingkan jika kesalahan ditemukan setelah aplikasi digunakan oleh publik. Selain itu, *unit testing* menjamin stabilitas sistem melalui proses regresi; ketika dilakukan penambahan fitur baru, pengujian otomatis akan memastikan bahwa perubahan tersebut tidak merusak fitur yang sudah ada sebelumnya.

2. **Mengapa branch coverage sulit mencapai 100%?**
   Branch coverage mengukur sejauh mana semua jalur percabangan logika (seperti if/else, switch case, atau *ternary operator*) telah dieksekusi oleh tes. Mencapai angka 100% sangat sulit karena beberapa faktor:
   - **Kompleksitas Logika Internal:** Banyak percabangan yang dihasilkan secara otomatis oleh *framework* (seperti Next.js) atau pustaka pihak ketiga yang sulit diakses langsung oleh pengembang.
   - **Defensive Programming:** Seringkali pengembang menuliskan blok else atau default hanya untuk mengantisipasi kondisi ekstrem yang secara praktis mustahil terjadi dalam simulasi pengujian.
   - **Keterbatasan Lingkungan Test:** Beberapa jalur logika hanya aktif pada lingkungan server (*Server-side*) atau browser tertentu, sehingga sulit bagi *test runner* seperti Jest untuk mencakup semua skenario tersebut secara bersamaan.

3. **Apa itu mocking?**
   Mocking adalah sebuah teknik simulasi dalam pengujian perangkat lunak yang bertujuan untuk mengganti ketergantungan (*dependency*) asli dengan objek tiruan. Dalam konteks Next.js, *mocking* sering diterapkan pada modul seperti next/router atau pustaka pengambilan data seperti SWR. Hal ini dilakukan agar unit test dapat berjalan secara terisolasi tanpa harus bergantung pada ketersediaan API eksternal, koneksi database, atau perilaku browser yang kompleks. Dengan *mocking*, pengujian dapat difokuskan sepenuhnya pada logika komponen yang sedang dites dalam kondisi yang terkontrol.

4. **Kapan snapshot test digunakan?**
   Snapshot test digunakan ketika ada kebutuhan untuk memantau konsistensi antarmuka pengguna (UI). Pengujian ini bekerja dengan cara mengambil "potret" struktur HTML yang dihasilkan oleh suatu komponen dan membandingkannya dengan hasil di masa mendatang. Metode ini sangat efektif digunakan pada komponen yang memiliki struktur UI stabil. Jika terjadi perubahan struktur HTML yang tidak disengaja—misalnya karena perubahan *styling* atau penghapusan elemen—maka pengujian akan memberikan peringatan, sehingga pengembang dapat meninjau apakah perubahan tersebut memang diinginkan atau merupakan sebuah kesalahan.

5. **Apakah semua file harus dites?**
   Secara teoritis, pengujian menyeluruh sangat ideal, namun dalam prakteknya tidak semua file wajib memiliki unit test. Pengujian harus diprioritaskan pada file yang mengandung logika bisnis yang kompleks (*logic-heavy*) dan fitur-fitur kritis (*critical features*) yang berdampak langsung pada fungsionalitas utama aplikasi. File yang bersifat statis, file konfigurasi (seperti jest.config.mjs), atau file yang hanya berisi definisi tipe data (seperti .d.ts) umumnya tidak memerlukan unit test. Target utama dalam pengembangan profesional adalah mencapai ambang batas *coverage* yang aman (umumnya di atas 80%) untuk memastikan keandalan aplikasi tanpa mengorbankan efisiensi waktu pengembangan.